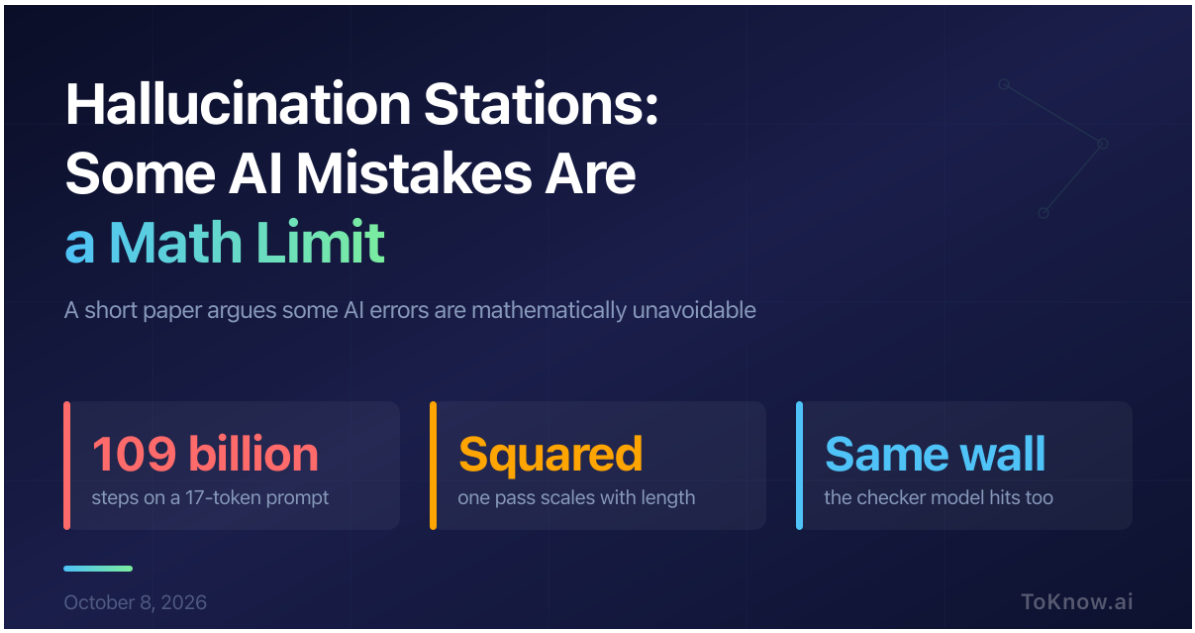
Hallucination Stations: Some AI Mistakes Are a Math Limit, Not a Bug

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Two researchers argue that some AI mistakes are not bugs but a hard ceiling on how much arithmetic a language model can do in one pass. Hallucination Stations is a short paper by Varin Sikka, a Stanford student, and his father Vishal Sikka, a former Infosys chief executive. The claim comes from how a model works: each pass costs about the square of the input length, so every prompt gets the same fixed computation. They measured one. A small Llama

model runs exactly 109,243,372,873 floating-point operations, the add-and-multiply steps a chip performs, on any 17-token prompt, whether it explains renewable energy or solves a puzzle. When a task needs more steps than that budget allows, the model cannot get it right.

Listing every combination of a set of tokens, or multiplying two large matrices, needs far more steps than one pass supplies. The paper states it as a theorem: if a prompt contains a task more complex than the model's fixed budget, the answer will be wrong, following a 1965 result that says some problems need more steps than others. That matters for scheduling, routing, software verification and simulation, where teams now hand an agent a problem and trust the number it returns.

Checking the work does not help: a second model hits the same ceiling, and verifying a hard answer can cost more than producing it. The bound covers one uninterrupted guess. A model that writes out a long chain of thought or runs an offline review pass spends more computation and can reach further, though the authors say those extra tokens are far fewer than the task needs.

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Sources:

- Hallucination Stations: On Some Basic Limitations of Transformer-Based Language Models (arXiv)
- AI Agents Are Poised to Hit a Mathematical Wall, Study Finds (Gizmodo)
- The Math on AI Agents Does Not Add Up (WIRED)
- The Illusion of Thinking (Apple Machine Learning Research)
- Chain of Thought Empowers Transformers to Solve Inherently Serial Problems (arXiv)

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